PRODUCT USER MANUAL

For Global Sea Ice Thickness Products

SEAICE_GLO_PHY_L4_NRT_011_014
Issue: 2.2

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	31.10.2019	All	Initial version	Jaakko Seppänen	C. Wettre
1.1	15.01.2021	II.5	Update changed from weekly to daily	Jaakko Seppänen	C. Wettre
1.2	16.02.2022	I.1, III.4, III.5	Updated product name from L3 to L4	Jaakko Seppänen	C. Wettre
2.0	28.05.2024	All	Update of the production system to version 3 of the source dataset, including the addition of data from the Southern Hemisphere.	Stefan Hendricks	C. Wettre
2.1	28.05.2025	All	Update of SMOS input data version in the northern hemisphere	Stefan Hendricks	C. Wettre
2.2	06.03.2025	2.a, 2.b, 2.c, 2.g, 7	Implementing consistent terminology for temporal coverage and resolution. Add OSI-SAF dataset DOI's, Minor editorial changes	Stefan Hendricks	C. Wettre

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
SI	Sea Ice
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
CIS	Central Information System
DU	Dissemination Unit. A Dissemination Unit is a kind of Production Unit, with the ability to disseminate its own products or other Production Unit's products to the rest of CMEMS Production Centres and to external Users.
DT	Delayed Time
EC	European Commission
ECMWF	European Centre for Medium Range Weather Forecast
SAR	Synthetic Aperture Radar
SIC	Sea Ice Concentration
SIT	Sea Ice Thickness
SIType	Sea Ice Type (first-year / multi-year sea ice)
TAC	Thematic Assembly Centre
SIT	Sea Ice Thickness
NetCDF	Network Common Data Form
CF	Climate and Forecast, format for metadata conventions

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document describes the data product SEAICE_ARC_PHY_L4_NRT_011_014 from the Copernicus Marine Service, what data services are available to access them, and how to use the files and services.

The product utilizes the measurements of CryoSat-2, Sentinel-3A, Sentinel-3B and Soil Moisture and Ocean Salinity (SMOS) satellites and an algorithm developed in the Alfred Wegener Institute. It contains winter sea ice thickness in both hemispheres, with one dataset generated per hemisphere in mutually exclusive periods. Both data sets are generated in near real-time with a latency of 2 days with respect to the end of the 7-day observation period of each file.

This product is produced by the Alfred Wegener Institute in the framework of the ESA CryoSat-2/SMOS data production and dissemination project and disseminated through the central Copernicus Marine Service dissemination unit by the Finnish Meteorological Institute with minor adaption of the file name.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SEAICE_GLO_PHY_L4_NRT_011_014
Geographical coverage	Northern hemisphere : 180°W-180°E ; 40°N – 90°N Southern hemisphere : 180°W-180°E ; 40°S – 90°S
Variables	• Sea Ice Thickness • Sea Ice Thickness Uncertainty • Sea Ice Concentration • Density of Ocean
Product Type	Near Real-Time
Available time series	-2Y to – 5 days
Temporal resolution	Weekly
Update Frequency	Daily
Target delivery time	daily at 12 pm (noon) UTC
Horizontal resolution	12.5 km grid spacing
Number of vertical levels	1
Format	NetCDF4

The scientific background can be found in the following references.

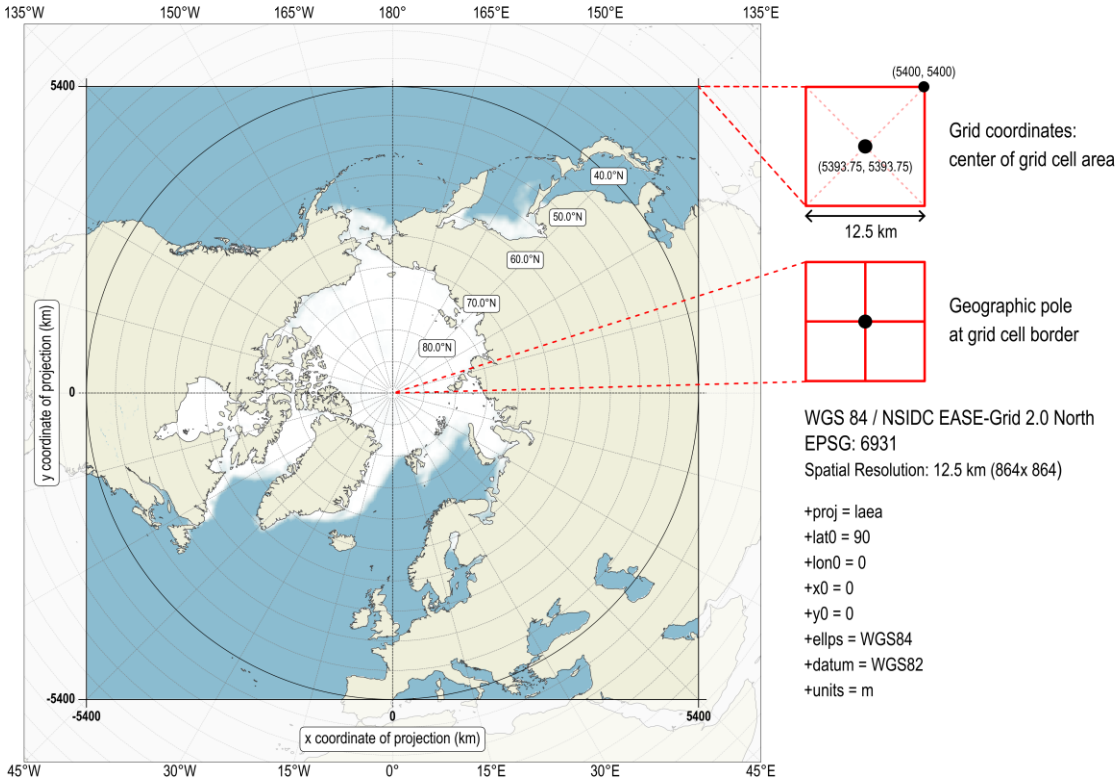
Topic	**Reference**
SMOS sea ice thickness retrieval	Tian-Kunze et al., 2014 (Northern hemisphere) Kaleschke et al. 2024 (Southern hemisphere)
CryoSat-2 sea ice thickness retrieval	Hendricks & Paul, 2023
Optimal Interpolation of CryoSat-2 & SMOS data	Ricker et al. 2017

b) Production System Description

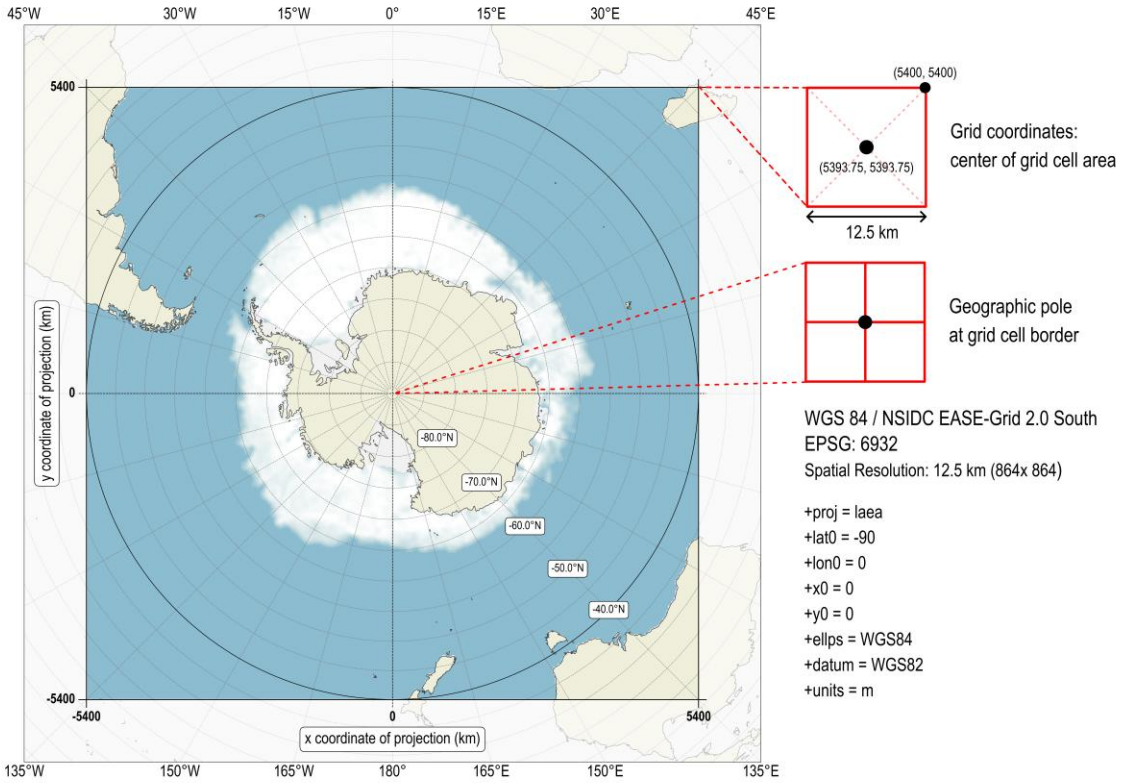
The production system has been developed and is operated at the Alfred Wegener Institute in the frame of the ESA CryoSat-2/SMOS production and dissemination service. The website[1] of the project provides additional information and documentation of the dataset.

GLOBAL (180°W-180°E; NH: 40°N-90°N, SH: 40°S-90°S): 12.5km, regular grid, 864 x 864

This product is global with one dataset per polar ocean in the northern and southern hemisphere. Each dataset is computed on a polar equal-area projection. One extent (10800 km) and spatial resolution (12.5km) is used for both hemispheres only the central latitude of the Lambert Azimuthal Equal Area projection is 90° for the northern hemisphere and -90° for the southern hemisphere respectively.

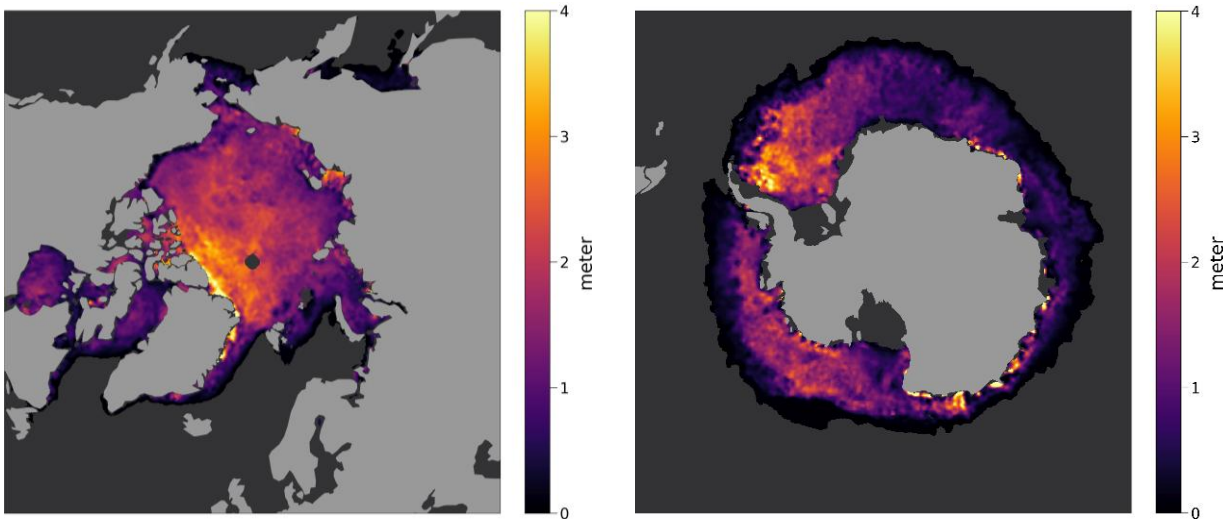


[1] https://earth.esa.int/eogateway/catalog/smos-cryosat-l4-sea-ice-thickness (last accessed May 2025)



Vertical grid	Surface only	
Version	v300	
Horizontal Resolution	12.5 km	
Production Unit	Alfred Wegener Institute, Germany	
Observations	SMOS MIRAL CryoSat-2 SIRAL Sentinel-3A SRAL Sentinel-3B SRAL	
Auxiliary Data Files	Sea Ice Concentration	OSI-401 (OSI-SAF, 2017) OSI-408 (OSI-SAF, 2017b) *after July 2025*
	Sea Ice Type	OSI-403 (OSI-SAF,2017a)
	Density of Ocean	ESA CCI surface type mask

Example of Northern Hemisphere (left) and Southern Hemispher sea ice thickness:



c) Dataset details

Product Lines	SEAICE_GLO_PHY_L4_NRT_011_014	
Geographical coverage	Northern hemisphere : 180°W-180°E ; 40°N – 90°N Southern hemisphere : 180°W-180°E ; 40°S – 90°S	
Datasets: esa_obs-si_arc_phy-sit_nrt_l4-multi_P1D-m esa_obs-si_ant_phy-sit_nrt_l4-multi_P1D-m		
Product Type	Near Real-Time	
Update frequency	Daily	
Available time series	-2Y - -5 days	
Target delivery time	Daily at 12:00	
Temporal resolution	Weekly	

The file structure of the two datasets for the northern and southern hemisphere is identical.

Datasets : **esa_obs-si_arc_phy-sit_nrt_l4-multi_P1D-m** **esa_obs-si_arc_phy-sit_nrt_l4-multi_P1D-m** contain 2D 7-da mean fields: sea ice thickness, sea ice thickness uncertainty, sea ice concentration, fractional land/ocean mask, status flag, quality flag
sea_ice_thickness [m] Sea Ice Thickness sea_ice_thickness
uncertainty [m] Sea Ice Thickness Uncertainty sea_ice_thickness standard_error
status_flag [1] Sea Ice Thickness Status Flag status_flag
quality_flag [1] Sea Ice Thickness Quality Flag quality_flag
sea_ice_concentration [%] Sea Ice Concentration sea_ice_area_fraction
density_of_ocean [1] Density_of_ocean sea_ice_area_fraction

d) Additional information on some parameters

sea_ice_thickness [m]	Mean sea ice thickness of the ice-covered area fraction of the grid cell. Derived by optimal interpolation of the Ku-Band radar altimeter and L-Band radiometer data.
sea_ice_thickness_uncertainty [m]	Algorithm uncertainty of the gridded data
sea_ice_concentration [%]	Sea ice concentration from auxiliary data averaged over the 7-day observation period.
density_of_ocean [1]	Ocean surface type area fraction per grid cell. Can be used for the computation of sea ice volume: area * density_of_ocean * sea_ice_concentration * sea_ice_thickness
Status_flag [1]	Flag indicating the status of the retrieval A flag indicating the status of the retrieval: 0: nominal_retrieval 1: open ocean 2: land, lake or Land ice 3: retrieval failed
quality_flag [1]	A flag indicating an available source data for the optimal interpolation at the grid cell: 0: altimetry radiometry 1: altimetry 2: radiometry 3: interpolated 4: outside sea ice mask

e) Processing information

Product disseminating schedule :

Type of product	Latency	Product temporal coverage (days)	Product disseminating scheduled time	Dissemination day
Near Real-Time	2 days	7	10am -> 12am	Every day

f) Catalogue updating

SEAICE_GLO_PHY_L4_NRT_011_014 are 7 day mean fields centered at 12:00 UTC og the 4th day. The product is updated daily for the winter periods in the respective hemisphere :

Hemisphere	Product Reference Day	Observation Period
Northern Hemisphere	Oct 18: 12:00 till Apr 12: 12:00	Oct 15 – Oct 21 till Apr. 08 – Apr 15
Southern Hemisphere	Apr 18: 12:00 till Oct 12: 12:00	Apr 15 – Apr 21 till Oct. 08 – Oct. 15

The latency of production is 2 days with respect to the end of the observation period. E.g. the last file of the winter period of the northern hemisphere (Apr. 08 – Apr 15) is delivered on April 17 12:00.

g) Time averaging

The result of each file consists of the optimal interpolation of satellite data within 7 calendar days in the UTC time zone.

Dataset frequency	Time averaging	Production Latency
daily	All satellite overpasses within a weekly period (UTC days) are used for observation field	2 days after last day of observation file

Daily auxiliary data files, such as sea ice concentration, are average over the same 7-day observation period.

h) Product use and impact

The 2002 implementation of plan of the Global Climate Observing System (GCOS) (WMO, 2022) specifically calls for action regarding the remote sensing of sea ice thickness and volume. The potential application of SEAICE_GLO_PHY_L4_NRT_011_014 are those where such information is required in near real-time, for example in the form of assimilation into numerical sea ice models, or sea ice assessements with short latency.

Topic	Example Reference
Assimilation into Arctic Reanalysis (TOPAZ4)	Xie et al., 2018
Assimilation into Sea Ice Forecasting Systems	Mignac et al. 2022
Sea Ice Volume as metric for for state of the marine cryosphere (NOAA Arctic Report Card)	Druckenmiller et al., 2024

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

5) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
esa_obs-si_arc_phy-sit_nrt_l4-multi_P1D-m	W_XX_ESA_SMOS_CS2_NH_12P5KM_EASE2_20230330_20230405_o_v300_01_l4sit.nc	0.01
esa_obs-si_ant_phy-sit_nrt_l4-multi_P1D-m	W_XX_ESA_SMOS_CS2_SH_12P5KM_EASE2_20250415_20250420_o_v300_01_l4sit.nc	0.01

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex,** for each dataset.

7) REFERENCES

Quality Information Document:
https://documentation.marine.copernicus.eu/QUID/CMEMS-SEAICE-QUID-011-014.pdf

OSI SAF (2017): Global Sea Ice Concentration (netCDF) - DMSP, EUMETSAT SAF on Ocean and Sea Ice, DOI: 10.15770/EUM_SAF_OSI_NRT_2004, https://doi.org/10.15770/EUM_SAF_OSI_NRT_2004

OSI SAF (2017a): Global Sea Ice Type (netCDF) - Multimission, EUMETSAT SAF on Ocean and Sea Ice, DOI: 10.15770/EUM_SAF_OSI_NRT_2006, https://doi.org/10.15770/EUM_SAF_OSI_NRT_2006

OSI SAF (2017b): Global AMSR Sea Ice Concentration - GCOM-W1, EUMETSAT SAF on Ocean and Sea Ice, DOI: 10.15770/EUM_SAF_OSI_NRT_2023, https://doi.org/10.15770/EUM_SAF_OSI_NRT_2023

Druckenmiller, M. L., and Coauthors, 2024: The Arctic. *Bull. Amer. Meteor. Soc.*, 105, S277–S330, https://doi.org/10.1175/BAMS-D-24-0101.1.

World Meteorological Organization (WMO), United Nations Educational, Scientific and Cultural Organization, Intergovernmental Oceanographic Commission, United Nations Environment Programme, International Science Council , (ISC) (2022): The 2022 GCOS Implementation Plan (GCOS-244), *https://library.wmo.int/idurl/4/58104*

Hendricks, S., & Paul, S. (2023). Product User Guide & Algorithm Specification - AWI CryoSat-2 Sea Ice Thickness (version 2.6) Issued by (v2.6). Zenodo. https://doi.org/10.5281/zenodo.10044554

Kaleschke, L., Tian-Kunze, X., Hendricks, S., and Ricker, R.: SMOS-derived Antarctic thin sea ice thickness: data description and validation in the Weddell Sea, Earth Syst. Sci. Data, 16, 3149–3170, https://doi.org/10.5194/essd-16-3149-2024, 2024..

Mignac, D., Martin, M., Fiedler, E., Blockley, E. & Fournier, N.(2022) Improving the Met Office's Forecast Ocean Assimilation Model (FOAM) with the assimilation of satellite-derived sea-ice thickness data from CryoSat-2 and SMOS in the Arctic. Q J R Meteorol Soc, 148(744), 1144–1167. Available from: https://doi.org/10.1002/qj.4252

Ricker, R., Hendricks, S., Kaleschke, L., Tian-Kunze, X., King, J., and Haas, C.: A weekly Arctic sea-ice thickness data record from merged CryoSat-2 and SMOS satellite data, The Cryosphere, 11, 1607–1623, https://doi.org/10.5194/tc-11-1607-2017, 2017.

Tian-Kunze, X., Kaleschke, L., Maaß, N., Mäkynen, M., Serra, N., Drusch, M., and Krumpen, T.: SMOS-derived thin sea ice thickness: algorithm baseline, product specifications and initial verification, The Cryosphere, 8, 997–1018, https://doi.org/10.5194/tc-8-997-2014, 2014

Xie, J., Counillon, F., and Bertino, L.: Impact of assimilating a merged sea-ice thickness from CryoSat-2 and SMOS in the Arctic reanalysis, The Cryosphere, 12, 3671–3691, https://doi.org/10.5194/tc-12-3671-2018, 2018.

8) ANNEX

a) Structure and semantic of NetCDF maps files

For dataset esa_obs-si_arc_phy-sit_nrt_l4-multi_P1D-m

```
dimensions:
  time = 1;
  xc = 864;
  yc = 864;
  time_bnds = 2;

variables:
  double time(time=1);
    :_FillValue = NaN; // double
    :long_name = "time";
    :units = "seconds since 1970-01-01";
    :calendar = "gregorian";
    :_ChunkSizes = 1U; // uint

  long Lambert_Azimuthal_Equal_Area;
    :semi_major_axis = 6378137.0; // double
    :semi_minor_axis = 6356752.314245179; // double
    :inverse_flattening = 298.257223563; // double
    :reference_ellipsoid_name = "WGS 84";
    :longitude_of_prime_meridian = 0.0; // double
    :prime_meridian_name = "Greenwich";
    :geographic_crs_name = "WGS 84";
    :horizontal_datum_name = "World Geodetic System 1984 ensemble";
    :projected_crs_name = "WGS 84 / NSIDC EASE-Grid 2.0 North";
    :grid_mapping_name = "lambert_azimuthal_equal_area";
    :latitude_of_projection_origin = 90.0; // double
    :longitude_of_projection_origin = 0.0; // double
    :false_easting = 0.0; // double
    :false_northing = 0.0; // double
    :proj4_str = "+proj=laea +lat_0=90 +lon_0=0 +x_0=0 +y_0=0 +datum=WGS84
        +units=m +no_defs +type=crs";

  double xc(xc=864);
    :_FillValue = NaN; // double
    :axis = "X";
    :units = "m";
    :long_name = "x coordinate in Cartesian system";
    :standard_name = "projection_x_coordinate";
    :_ChunkSizes = 864U; // uint

  double yc(yc=864);
    :_FillValue = NaN; // double
    :axis = "Y";
    :units = "m";
    :long_name = "y coordinate in Cartesian system";
```

```
    :standard_name = "projection_y_coordinate";
    :_ChunkSizes = 864U; // uint

  double lon(yc=864, xc=864);
    :_FillValue = NaN; // double
    :units = "degrees_east";
    :long_name = "longitude coordinate";
    :standard_name = "longitude";
    :_ChunkSizes = 864U, 864U; // uint

  double lat(yc=864, xc=864);
    :_FillValue = NaN; // double
    :units = "degrees_north";
    :long_name = "latitude coordinate";
    :standard_name = "latitude";
    :_ChunkSizes = 864U, 864U; // uint

  double sea_ice_thickness(time=1, yc=864, xc=864);
    :_FillValue = NaN; // double
    :long_name = "sea ice thickness";
    :standard_name = "sea_ice_thickness";
    :units = "m";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :ancillary_variables = "sea_ice_thickness_uncertainty status_flag
        quality_flag";
    :comment = "derived by optimal interpolation";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  double sea_ice_thickness_uncertainty(time=1, yc=864, xc=864);
    :_FillValue = NaN; // double
    :long_name = "sea ice thickness uncertainty";
    :standard_name = "sea_ice_thickness standard_error";
    :units = "m";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :comment = "derived by optimal interpolation";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  float sea_ice_concentration(time=1, yc=864, xc=864);
    :_FillValue = NaNf; // float
    :long_name = "sea ice concentration";
    :standard_name = "sea_ice_area_fraction";
    :units = "percent";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  float density_of_ocean(yc=864, xc=864);
    :_FillValue = NaNf; // float
    :long_name = "density of ocean";
    :units = "1";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
```

```
    :comment = "Source: ESA CCI sea ice project (credit: Thomas Lavergne,
        MET Norway)";
    :coordinates = "lat lon";
    :_ChunkSizes = 864U, 864U; // uint

  int status_flag(time=1, yc=864, xc=864);
    :long_name = "Sea Ice Thickness Status Flag";
    :standard_name = "status_flag";
    :units = "1";
    :valid_min = 0; // int
    :valid_max = 3; // int
    :flag_values = 0, 1, 2, 3; // int
    :flag_meanings = "nominal_retrieval open_ocean land_lake_landice
        retrieval_failed";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  int quality_flag(time=1, yc=864, xc=864);
    :long_name = "Sea Ice Thickness Quality Flag";
    :standard_name = "quality_flag";
    :units = "1";
    :valid_min = 0; // int
    :valid_max = 4; // int
    :flag_values = 0, 1, 2, 3, 4; // int
    :flag_meanings = "altimetry_radiometry altimetry radiometry
        interpolated no_sea_ice";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  double time_bnds(time_bnds=2);
    :_FillValue = NaN; // double
    :long_name = "time_bnds";
    :units = "seconds since 1970-01-01";
    :calendar = "gregorian";
    :_ChunkSizes = 2U; // uint

// global attributes:
:title = "Sea Ice Thickness derived from merging CryoSat-2, Sentinel-3A,
        Sentinel-3B and SMOS ice thickness";
:description = "Arctic sea-ice thickness derived from CryoSat-2 and SMOS
        using an optimal interpolation scheme";
:summary = "Arctic sea-ice thickness derived from CryoSat-2, Sentinel-3A,
        Sentinel-3B and SMOS using an optimal interpolation scheme";
:keywords = "Cryosphere > Sea Ice > Sea Ice Thickness";
:product_version = "300";
:processing_mode = "o";
:time_of_creation = "2025-04-06T10:06:17.487611";
:history = "2025-04-06T10:06:17.487611 creation";
:cdm_data_type = "Grid";
:Conventions = "CF-1.6";
:spatial_resolution = "12.5 km grid spacing";
```

```
  :geospatial_lat_min = 89.92086560475803; // double
  :geospatial_lat_max = 89.92086560475803; // double
  :geospatial_lon_min = -180.0; // double
  :geospatial_lon_max = 180.0; // double
  :geospatial_vertical_min = 0.0; // double
  :geospatial_vertical_max = 0.0; // double
  :geospatial_bounds_crs = "EPSG:6931";
  :time_coverage_start = "2025-03-29 00:00:00+00:00";
  :time_coverage_end = "2025-04-04 23:59:59.999999+00:00";
  :time_coverage_duration = "P7D";
  :time_coverage_resolution = "P1D";
  :platform = "CryoSat-2, Sentinel-3A, Sentinel-3B, SMOS";
  :sensor = "SIRAL, SRAL, SRAL, MIRAS";
  :project = "CS2SMOS PDS: SMOS & CryoSat-2 Sea Ice Data Product Processing
          and Dissemination Service";
  :institution = "Alfred-Wegener-Institut Helmholtz Zentrum für Polar und
          Meeresforschung (AWI)";
  :creator_name = "Alfred-Wegener-Institut Helmholtz Zentrum für Polar und
          Meeresforschung (AWI)";
  :creator_type = "institution";
  :creator_url = "www.awi.de";
  :contributor_name = "Stefan Hendricks, Xiangshan Tian-Kunze, Lars
          Kaleschke, Robert Ricker";
  :contributor_role = "PrincipalInvestigator, Author, Author, Author";
  :publisher_email = "cs2smos-support@awi.de";
  :publisher_url = "https://spaces.awi.de/confluence/x/DwVmEQ";
  :references = "Ricker, R., Hendricks, S., Kaleschke, L., Tian-Kunze, X.,
          King, J., and Haas, C.: A weekly Arctic sea-ice thickness data
          record from merged CryoSat-2 and SMOS satellite data, The
          Cryosphere, 11, 1607-1623, https://doi.org/10.5194/tc-11-1607-
          2017, 2017.";
  :coordinates = "Lambert_Azimuthal_Equal_Area";
}
```

For dataset esa_obs-si_ant_phy-sit_nrt_l4-multi_P1D-m

```
dimensions:
    time = 1;
    xc = 864;
    yc = 864;
    time_bnds = 2;

variables:
    double time(time=1);
      :_FillValue = NaN; // double
      :long_name = "time";
      :units = "seconds since 1970-01-01";
      :calendar = "gregorian";
      :_ChunkSizes = 1U; // uint

    long Lambert_Azimuthal_Equal_Area;
```

```
    :semi_major_axis = 6378137.0; // double
    :semi_minor_axis = 6356752.314245179; // double
    :inverse_flattening = 298.257223563; // double
    :reference_ellipsoid_name = "WGS 84";
    :longitude_of_prime_meridian = 0.0; // double
    :prime_meridian_name = "Greenwich";
    :geographic_crs_name = "WGS 84";
    :horizontal_datum_name = "World Geodetic System 1984 ensemble";
    :projected_crs_name = "WGS 84 / NSIDC EASE-Grid 2.0 South";
    :grid_mapping_name = "lambert_azimuthal_equal_area";
    :latitude_of_projection_origin = -90.0; // double
    :longitude_of_projection_origin = 0.0; // double
    :false_easting = 0.0; // double
    :false_northing = 0.0; // double
    :proj4_str = "+proj=laea +lat_0=-90 +lon_0=0 +x_0=0 +y_0=0 +datum=WGS84
        +units=m +no_defs +type=crs";

  double xc(xc=864);
    :_FillValue = NaN; // double
    :axis = "X";
    :units = "m";
    :long_name = "x coordinate in Cartesian system";
    :standard_name = "projection_x_coordinate";
    :_ChunkSizes = 864U; // uint

  double yc(yc=864);
    :_FillValue = NaN; // double
    :axis = "Y";
    :units = "m";
    :long_name = "y coordinate in Cartesian system";
    :standard_name = "projection_y_coordinate";
    :_ChunkSizes = 864U; // uint

  double lon(yc=864, xc=864);
    :_FillValue = NaN; // double
    :units = "degrees_east";
    :long_name = "longitude coordinate";
    :standard_name = "longitude";
    :_ChunkSizes = 864U, 864U; // uint

  double lat(yc=864, xc=864);
    :_FillValue = NaN; // double
    :units = "degrees_north";
    :long_name = "latitude coordinate";
    :standard_name = "latitude";
    :_ChunkSizes = 864U, 864U; // uint

  double sea_ice_thickness(time=1, yc=864, xc=864);
    :_FillValue = NaN; // double
    :long_name = "sea ice thickness";
    :standard_name = "sea_ice_thickness";
    :units = "m";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
```

```
    :ancillary_variables = "sea_ice_thickness_uncertainty status_flag
        quality_flag";
    :comment = "derived by optimal interpolation";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  double sea_ice_thickness_uncertainty(time=1, yc=864, xc=864);
    :_FillValue = NaN; // double
    :long_name = "sea ice thickness uncertainty";
    :standard_name = "sea_ice_thickness standard_error";
    :units = "m";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :comment = "derived by optimal interpolation";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  float sea_ice_concentration(time=1, yc=864, xc=864);
    :_FillValue = NaNf; // float
    :long_name = "sea ice concentration";
    :standard_name = "sea_ice_area_fraction";
    :units = "percent";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  float density_of_ocean(yc=864, xc=864);
    :_FillValue = NaNf; // float
    :long_name = "density of ocean";
    :units = "1";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :comment = "Source: ESA CCI sea ice project (credit: Thomas Lavergne,
        MET Norway)";
    :coordinates = "lat lon";
    :_ChunkSizes = 864U, 864U; // uint

  int status_flag(time=1, yc=864, xc=864);
    :long_name = "Sea Ice Thickness Status Flag";
    :standard_name = "status_flag";
    :units = "1";
    :valid_min = 0; // int
    :valid_max = 3; // int
    :flag_values = 0, 1, 2, 3; // int
    :flag_meanings = "nominal_retrieval open_ocean land_lake_landice
        retrieval_failed";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  int quality_flag(time=1, yc=864, xc=864);
    :long_name = "Sea Ice Thickness Quality Flag";
    :standard_name = "quality_flag";
    :units = "1";
    :valid_min = 0; // int
```

```
    :valid_max = 4; // int
    :flag_values = 0, 1, 2, 3, 4; // int
    :flag_meanings = "altimetry_radiometry altimetry radiometry
        interpolated no_sea_ice";
    :grid_mapping = "Lambert_Azimuthal_Equal_Area";
    :coordinates = "time lat lon";
    :_ChunkSizes = 1U, 864U, 864U; // uint

  double time_bnds(time_bnds=2);
    :_FillValue = NaN; // double
    :long_name = "time_bnds";
    :units = "seconds since 1970-01-01";
    :calendar = "gregorian";
    :_ChunkSizes = 2U; // uint

// global attributes:
:title = "Sea Ice Thickness derived from merging CryoSat-2, Sentinel-3A,
        Sentinel-3B and SMOS ice thickness";
:description = "Antarctic sea-ice thickness derived from CryoSat-2 and SMOS
        using an optimal interpolation scheme";
:summary = "Antarctic sea-ice thickness derived from CryoSat-2, Sentinel-
        3A, Sentinel-3B and SMOS using an optimal interpolation scheme";
:keywords = "Cryosphere > Sea Ice > Sea Ice Thickness";
:product_version = "300";
:processing_mode = "o";
:time_of_creation = "2025-04-22T11:52:25.382988";
:history = "2025-04-22T11:52:25.382988 creation";
:cdm_data_type = "Grid";
:Conventions = "CF-1.6";
:spatial_resolution = "12.5 km grid spacing";
:geospatial_lat_min = -89.92086560475803; // double
:geospatial_lat_max = -16.524349963478976; // double
:geospatial_lon_min = -180.0; // double
:geospatial_lon_max = 180.0; // double
:geospatial_vertical_min = 0.0; // double
:geospatial_vertical_max = 0.0; // double
:geospatial_bounds_crs = "EPSG:6932";
:time_coverage_start = "2025-04-14 00:00:00+00:00";
:time_coverage_end = "2025-04-20 23:59:59.999999+00:00";
:time_coverage_duration = "P7D";
:time_coverage_resolution = "P1D";
:platform = "CryoSat-2, Sentinel-3A, Sentinel-3B, SMOS";
:sensor = "SIRAL, SRAL, SRAL, MIRAS";
:project = "CS2SMOS PDS: SMOS & CryoSat-2 Sea Ice Data Product Processing
        and Dissemination Service";
:institution = "Alfred-Wegener-Institut Helmholtz Zentrum für Polar und
        Meeresforschung (AWI)";
:creator_name = "Alfred-Wegener-Institut Helmholtz Zentrum für Polar und
        Meeresforschung (AWI)";
:creator_type = "institution";
:creator_url = "www.awi.de";
:contributor_name = "Stefan Hendricks, Xiangshan Tian-Kunze, Lars
        Kaleschke, Robert Ricker";
```

```
  :contributor_role = "PrincipalInvestigator, Author, Author, Author";
  :publisher_email = "cs2smos-support@awi.de";
  :publisher_url = "https://spaces.awi.de/confluence/x/DwVmEQ";
  :references = "Ricker, R., Hendricks, S., Kaleschke, L., Tian-Kunze, X.,
          King, J., and Haas, C.: A weekly Arctic sea-ice thickness data
          record from merged CryoSat-2 and SMOS satellite data, The
          Cryosphere, 11, 1607-1623, https://doi.org/10.5194/tc-11-1607-
          2017, 2017.";
  :coordinates = "Lambert_Azimuthal_Equal_Area";
}
```